SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I ____________vanumu surya satvika(192524254)________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:v.surya satvika

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Question 1

Constraint-Based Problem Solving using Backtracking Search

Given
Doctors:

- D1
- D2
- D3

Shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work Night.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor must be assigned exactly one shift.

Step 1: Initial Assignment
Available shifts:

- D1 → Morning, Afternoon
- D2 → Morning, Afternoon, Night
- D3 → Afternoon, Night

Step 2: Assign D1
Choose Morning.
Current Assignment
D1 → Morning
Remaining
Afternoon, Night
Constraint Check
✓ D1 is not assigned Night.

Step 3: Assign D2
Try Afternoon.
Current Assignment
D1 → Morning
D2 → Afternoon
Remaining
Night
Constraint Check
✓ D2 works before D3.

Step 4: Assign D3
Assign Night.
Current Assignment
D1 → Morning

D2 → Afternoon
D3 → Night
Constraint Check
✓ D3 is not Morning
✓ D2 before D3
✓ One doctor per shift
✓ Every doctor assigned once
All constraints satisfied.

Final Valid Schedule

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	Night

Backtracking Tree

```
Start
│
├── D1 → Morning ✓
│    │
│    ├── D2 → Afternoon ✓
│    │    │
│    │    └── D3 → Night ✓
│    │
│    └── Solution Found
```

Conclusion
Backtracking Search assigns doctors one by one while checking every constraint. Invalid assignments are rejected immediately, reducing unnecessary search. The final schedule satisfies all constraints.

2) BFS Search using Manhattan Distance

Given
Grid Size = **5 × 5**
Movement

- Up
- Down
- Left
- Right

Cost per move = **1**
Obstacle cells cannot be crossed.
Heuristic = Manhattan Distance.

Example Grid

```
S . . X .
. X . X .
```

. X . . .
. . X X .
. . . . G
S = Start
G = Goal
X = Obstacle

BFS Node Expansion
Step 1
Queue
[S]
Expand
S
Visited
S

Step 2
Neighbours
Right
Down
Queue
[(0,1),(1,0)]

Step 3
Expand (0,1)
Queue
[(1,0),(0,2)]

Step 4
Expand (1,0)
Queue
[(0,2),(2,0)]

Continue
Expand nodes level by level.
Ignore blocked cells.
Ignore visited nodes.

Manhattan Distance Formula
[
h(n)=|x_1-x_2|+|y_1-y_2|
]
Example
Current
(2,2)
Goal
(4,4)
[
h=|2-4|+|2-4|
]

=2+2
=4

Optimal Path
S
↓

↓

→

→

↓

↓

→

G

Total Cost
Number of moves = **8**
Cost = **8** × **1** = **8**

Conclusion
BFS explores nodes level by level and guarantees the shortest path when every move has equal cost.

3)(a) Constraint Analysis

The rescue robot works in a partially observable environment.
Constraints and Effects
Obstacle Cells
- Robot cannot pass through blocked rooms.
- It must search for an alternative path.

Limited Sensing
- Robot observes only adjacent rooms.
- It updates information while moving.

Risk Zones
- Risky cells increase movement cost.
- Robot avoids them whenever possible.

Movement Cost
- Every move costs 1.
- Safe paths with lower total cost are preferred.

Dynamic Environment
- The robot continuously updates its map.
- Decisions are made during execution.

(b) Solution Using Uniform Cost Search (UCS)

Step 1
Start from S.
Step 2
Store S in the priority queue.
Step 3
Expand the node with the lowest cumulative cost.
Step 4
Generate neighbouring nodes.
Step 5
Ignore blocked cells.
Step 6
Calculate movement cost.
Safe cell = +1
Risky cell = +3
Step 7
Insert updated nodes into the priority queue.
Step 8
Repeat until Goal (G) is reached.

UCS Flow
Start

↓

Expand Lowest Cost Node

↓

Generate Neighbours

↓

Remove Obstacles

↓

Update Cost

↓

Goal?

↓

Yes → Stop

No → Continue

(c) AI Concepts Applied

Intelligent Agent
The robot senses its surroundings using sensors and performs actions using actuators.
Rationality
The robot always chooses the safest and least costly path.
Environment Type

- Partially Observable
- Dynamic
- Sequential
- Deterministic (movement rules are fixed)

(d) Justification

Uniform Cost Search is the best choice because:

- Finds the minimum-cost path.
- Handles risky zones with different costs.
- Works effectively in dynamic environments.
- Guarantees an optimal solution when costs are non-negative.
- Suitable for online decision-making because the robot can update knowledge as new information is discovered.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided